JEONGRAK LEE

Postdoctoral Researcher
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Career Highlights

- **Sejong Science Fellowship** 2026–2031 (KRW 500,000,000)
- **14 first-author SCI papers** in the past 3 years (Top venues: *AIAA Journal, Aerospace Science and Technology, Fuel, Lab on a Chip, Journal of the Mechanics and Physics of Solids, International Journal of Mechanical Sciences*)
- **Co-founder, Perigee Aerospace Inc.** (2016–2019); led propulsion system development for the small orbital launch vehicle Blue Whale

EDUCATION

Pohang University of Science and Technology (POSTECH) — Pohang, Korea
Ph.D. *in Mechanical Engineering, Extreme Mechanics Lab*
Advisor: Prof. Anna Lee — Mar. 2020 – Feb. 2026
GPA: 4.13 / 4.30 (98.3 / 100)
Dissertation: Actuation Framework for Space Exploration: From Origami-Based Active Structures to In-Space Propulsion
Honors: Selected for Forbes 30 Under 30 Korea – Science, SBS Research Fellowship (KRW 97,000,000)

Pohang University of Science and Technology (POSTECH) — Pohang, Korea
B.S. *in Mechanical Engineering, Magna Cum Laude* — Mar. 2015 – Feb. 2020
Total GPA: 3.61 / 4.30 (93.1 / 100); Major GPA: 3.81 / 4.30 (95.1 / 100)
Honors: Mueunjae Award (POSTECH highest graduation honor); Next-Generation Engineering Leader Award (Highest Honor, National Academy of Engineering of Korea, NAEK); Talent Award of Korea (Minister of Education)

Daegu Il Science High School — Daegu, Korea
Mar. 2013 – Feb. 2015

RESEARCH INTERESTS

My research charts the path to **next-generation space propulsion**: from advanced chemical propulsion deployable today, to electrochemical hybrid propulsion for the near future, and ultimately to photonic propulsion that opens the deep-space exploration.

1. Advanced Chemical Propulsion (Mission-Ready)

(i) MEMS-based solid and liquid monopropellant micro-thrusters (50 mN – 1 N) and Lab-on-a-Chip propellant validation platforms for CubeSats and constellations; and (ii) plasma-assisted propulsion for small-scale (~ 1 N) mono-/bi-propellant thrusters and reliable ignition source of large-scale (> 100 N) engines using green storable propellants (N_2O/CH_4, N_2O/NH_3).

2. Chemical–Plasma Hybrid Propulsion (Near-Term Frontier)

(i) Novel Rotating Gliding Arc (RGA)-based plasma-assisted in-space propulsion systems for high-thrust, high-specific-impulse (5 – 22 N, > 500 s) orbital transfer vehicles and kick stages; (ii) AI-driven alloy development and additive manufacturing process optimization for RGA-compatible propulsion components; and (iii) multiphysics plasma–thermofluid modeling for thruster-geometry optimization.

3. Photonics-Based Space Propulsion (Long-Term Vision)

Next-generation light/solar sails enabled by integrating active-structure thin-film actuators with optical metasurfaces – generating direct photonic forces and reconfigurable attitude control for deep-space missions, supported by predictive mechanics of thin-film instabilities (wrinkling, buckling, large deformation).

· Other Aerospace Applications

Plasma-based high-enthalpy ground-test systems, including hypersonic and re-entry arc wind tunnels, ramjet operating environment simulation platform, fully electric jet engines, and VLEO atmosphere-breathing electric propulsion.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher Mar. 2026 – Present
Center for Advanced Aerospace Materials, Graduate Institute of Ferrous and Eco-Materials Technology (GIFT), POSTECH (alternative military service)

Graduate Researcher (Integrated M.S.–Ph.D. Program) Mar. 2020 – Feb. 2026
Extreme Mechanics Lab, Department of Mechanical Engineering, POSTECH (Advisor: Prof. Anna Lee)

Visiting Researcher Jul. 2024 – Sep. 2024
Plasma Engineering Laboratory, Korea Institute of Machinery & Materials (KIMM) (Host: Dr. Hongjae Kang)

Project-based Intern, Product Strategy Team Jan. 2021 – Feb. 2022
Hyundai Motor Group (ICT Innovation Talent 4.0 program, Ministry of Science and ICT)
- Developed an all-in-one EV vehicle / charger / service-platform commercialization model.

Undergraduate Researcher Jun. 2017 – Feb. 2018
Rocket Lab, Department of Aerospace Engineering, KAIST (Advisor: Prof. Sejin Kwon)
- Design, fabrication and testing of a green monopropellant thruster/reactor (premixed high-test peroxide).

Co-Founder & Propulsion Engineer Jul. 2016 – Apr. 2019
Perigee Aerospace Inc. / Perigee Rocket L.L.C.
- Small-satellite launch vehicle (Blue Whale) – high-altitude second-stage engine system integration.
- Contributed to TIPS (KRW 700 M), seed investment, and Series A funding.

INTERNATIONAL JOURNAL PUBLICATIONS

Published / Accepted (SCIE)
** Equal contribution; † Corresponding author*

Advanced Chemical Propulsion

1. **Lee, J.**, Kim, S., Jo, H., Lee, A.†, & Kang, H.† (2026). Plasma-assisted N_2O/CH_4 combustion in a rotating gliding arc thruster under ultra-lean conditions (O/F = 5–1000). ***Aerospace Science and Technology*** https://doi.org/10.1016/j.ast.2025.111256
2. **Lee, J.**, Kim, S., Jo, H., Lee, A.†, & Kang, H.† (2026). Ammonia-based storable bipropellant (N_2O/NH_3) thruster utilizing dual-mode plasma ignition for enhanced propellant flexibility. ***Fuel*** https://doi.org/10.1016/j.fuel.2025.136893
3. **Lee, J.***, Kim, S.*, Jo, H., & Lee, A.† (2024). Lab-on-PCB solid propellant microthruster with multi-mode thrust capabilities. ***Lab on a Chip*** [Selected as inside front cover] https://doi.org/10.1039/D4LC00157E
4. **Lee, J.**, Kim, S., Song, Y., Lee, S., Kang, H.†, & Lee, A.† (2024). Lab-on-PCB technology for liquid monopropellant microthrusters: design, fabrication, and performance evaluation. *Sensors and Actuators A: Physical* https://doi.org/10.1016/j.sna.2024.115347
5. **Lee, J.**, Kim, S., Ryoo, J., Kang, H.†, & Lee, A.† (2024). Conceptual demonstration of a Lego-like modular fabrication method for an engineering model of a small monopropellant PCB thruster. ***Acta Astronautica*** https://doi.org/10.1016/j.actaastro.2024.03.012
6. **Lee, J.***, Kim, S.*, Jo, H., Lee, E., Song, Y., & Lee, A.† (2023). Design and fabrication of a scalable solid-propellant micro-thruster array using Lab-on-PCB technology. *Sensors and Actuators A: Physical* https://doi.org/10.1016/j.sna.2023.114738
7. **Lee, J.***, Jo, H.*, Kim, S., Lee, E., Son, Y., & Lee, A.† (2023). Lab-on-PCB for space propulsion: integrated membraneless micro-ignitor for MEMS solid propellant thruster. *Sensors and Actuators A: Physical* https://doi.org/10.1016/j.sna.2023.114696
8. Kang, H., Kim, J., **Lee, J.**, & Kwon, S.† (2019). Green energetic monopropellant with a mixture of hydrogen peroxide and tetraglyme. ***Combustion and Flame*** https://doi.org/10.1016/j.combustflame.2019.10.007

Chemical–Plasma Hybrid Propulsion

9. **Lee, J.***, Kim, R.E.*, Lee, J.*, Lee, J., Kim, S., Kim, J., Kang, D., Kwon, Y., Kim, D.†, Lee, A.†, & Kim, H.S.† (2026). Surface roughness engineering of mechanical interlocking for enhanced metal–polymer adhesion in additive manufacturing. ***Virtual and Physical Prototyping*** https://doi.org/10.1080/17452759.2026.2653924
10. **Lee, J.**, Kim, S., Lee, A.†, & Kang, H.† (2026). Novel concept of a Newton-scale air-breathing plasma thruster using rotating arc discharge. *Advances in Space Research* https://doi.org/10.1016/j.asr.2026.01.005

11. **Lee, J.***, Lee, J.*, Kim, R.E.*, Kim, S., Kim, K., Kwon, Y., Kang, H., Yun, G., Kim, D.†, Lee, A.†, & Kim, H.S.† (2025). Architecture of Ti-6Al-4V thin-walled microthrusters fabricated via laser powder bed fusion. ***Virtual and Physical Prototyping*** https://doi.org/10.1080/17452759.2025.2533944
12. **Lee, J.**, Kim, J.W., Kim, S., Lee, A.†, & Kang, H.† (2025). Conceptual demonstration of hydrogen peroxide based electro-chemical propulsion with rotating gliding arc. ***AIAA Journal*** https://doi.org/10.2514/1.J064284
13. **Lee, J.**, Kim, S., Lee, A.†, & Kang, H.† (2025). An experimental study of performance and thrust control characteristics of N_2O-based rotating gliding arc thrusters. *Advances in Space Research* https://doi.org/10.1016/j.asr.2025.01.063
14. Kang, H., **Lee, J.**, Kim, K.-T., Song, Y., & Lee, D.† (2022). Conceptual demonstration of Martian atmosphere-breathing electrical supersonic thruster with CO_2-based rotating gliding arc. ***Acta Astronautica*** https://doi.org/10.1016/j.actaastro.2022.07.058

Photonics-Based Space Propulsion & Thin-Film Mechanics

15. Kwak, H.*, **Lee, J.***, Kim, J.*, Jo, H., Ki, K., & Lee, A.† (2026). Wrinkling of thin films in pre-stretched soft substrate trilayers. ***International Journal of Mechanical Sciences*** https://doi.org/10.1016/j.ijmecsci.2025.111213
16. Ki, K.*, **Lee, J.***, & Lee, A.† (2024). Combined influence of shallowness and geometric imperfection on the buckling of clamped spherical shells. ***Journal of the Mechanics and Physics of Solids*** https://doi.org/10.1016/j.jmps.2024.105554

Under Review

1. Kim, R.E.*, Lee, J.*, Ha, S.V., Kim, D., Lee, A., **Lee, J.**†, & Kim, H.S.†. Microstructure and mechanical properties of 316L stainless steel fabricated by laser wire directed energy deposition. *under review*.
2. **Lee, J.**, Lee, G., Kim, S., Jo, H., Lee, A.†, & Kang, H.†. Experimental investigation of swirl-injected bypass flow effects in an N_2O-based rotating gliding arc thruster. *under review*.
3. Kim, S.*, **Lee, J.***, Ki, K., Jo, H., & Lee, A.†. Multi-degree-of-freedom pouch motors enabled by integration of electrical and pneumatic systems using flexible PCB architecture. *under review*.
4. Kim, R.E.*, Lee, J.*, **Lee, J.***, Kim, S., Lee, J., Kim, K., Kwon, Y., Kang, H., Yun, G., Lee, A.†, Kim, D.†, & Kim, H.S.†. Additively manufactured structure-optimized Ti-6Al-4V microthrusters: unveiling size effects from sample to system level. *under review*.
5. **Lee, J.**, Kwak, H., Jo, H., & Lee, A.†. Thickness control and uniformity in dip-oscillation coating of cylindrical substrates. *under review*.
6. **Lee, J.**, Kim, S., Jo, H., Jung, J., Lee, A.†, & Kang, H.†. Manufacturing method for green monopropellant microthruster using metal PCB production process. *under review*.

CONFERENCE PRESENTATIONS

International Conferences

1. **Lee, J.**, & Lee, A. (2026). Ti–6Al–4V Thin-Walled Monopropellant Microthruster Designs Manufactured by Laser Powder Bed Fusion. ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM), June 8–10, Long Beach, CA, USA.
2. Lee, J., **Lee, J.,** & Lee, A. (2026). Enhancing Metal–polymer Adhesion via Surface Roughness Control in Laser Powder Bed Fusion for Space Structure Applications. ASME Aerospace Structures, Structural Dynamics, and Materials Conference (SSDM), June 8–10, Long Beach, CA, USA.
3. **Lee, J.**, Kim, S., Kang, H., & Lee, A. (2026). A Newton-scale air-breathing electric propulsion system based on rotating gliding arc. Int. Conf. Precision Engineering and Sustainable Manufacturing (PRESM), Singapore.
4. **Lee, J.**, Lee, J., Kim, R.E., Kim, S., & Lee, A. (2025). Design and characterization of thin-walled Ti-6Al-4V microthrusters fabricated by laser powder bed fusion. Int. Conf. Precision Engineering and Sustainable Manufacturing (PRESM), Chiang Mai, Thailand. **[Outstanding Presentation Award]**
5. **Lee, J.**, Kwak, H., Kim, J., Jo, H., & Lee, A. (2025). Wrinkling behavior of thin films encapsulated between two identical pre-stretched soft substrates. APS March Meeting, Anaheim, USA.
6. Kwak, H., **Lee, J.,** Lee, J., Kim, J., & Lee, A. (2024). Wrinkling of a thin film encapsulated between identically pre-stretched two soft substrates. Int. Congress of Theoretical and Applied Mechanics (ICTAM), Daegu, Korea.
7. Kim, S., **Lee, J.,** Lee, J., & Lee, A. (2024). Design of a scalable soft robot using thermo-pneumatic actuation on flexible electronics. IUTAM Symp. on Mechanics of Soft Materials and Soft Robots, Tokyo, Japan.
8. Kwak, H., **Lee, J.,** Lee, J., & Lee, A. (2023). Wrinkling of thin films between two soft layers for stretchable circuits with high packing density. Int. Conf. Precision Engineering and Sustainable Manufacturing (PRESM), Tokyo, Japan.

9. **Lee, J.**, Kwak, H., Jo, H., & Lee, A. (2023). Formation of wrinkled film layer between symmetrically stretched substrates. APS March Meeting, Las Vegas, USA.
10. **Lee, J.**, Kwak, H., Lee, B., Song, Y., & Lee, A. (2022). Uniform elastomer suspension coating on cylinders by bi-axial oscillation. Int. Conf. Precision Engineering and Sustainable Manufacturing (PRESM), Jeju, Korea.
11. Ki, K., **Lee, J.,** Lee, J., & Lee, A. (2022). From shallow to deep: buckling behavior of clamped spherical caps. APS March Meeting, Chicago, USA.
12. **Lee, J.**, Kwak, H., Lee, B., & Lee, A. (2021). Fabrication of films on cylinders by dip-oscillate coating. APS March Meeting (Online).
13. Ki, K., **Lee, J.**, & Lee, A. (2020). Shallowness effect on buckling of spherical shells. APS March Meeting (Online).

Domestic Conferences

30+ presentations at the Korean Society of Mechanical Engineers (KSME), the Korean Society for Aeronautical and Space Sciences (KSAS), the Korean Society for Precision Engineering, and the Korean Society of Propulsion Engineers – including three award-winning talks: **KSME 2022 Excellence Paper Award**, **Korean Society for Precision Engineering 2022 Best Paper Award**, and **KSME 2020 CAE/Applied Mechanics Best Paper Award**.

PATENTS

International Patents (3)

1. **Lee, J.** (first inventor), Jo, H., Lee, E., Son, Y., & Lee, Y. (2024). Solar panel module and spacecraft. U.S. Patent **No. US12012235**, registered 2024-06-18.
2. **Lee, J.** (first inventor), Lee, A., Lee, E., Son, Y., Jo, H., & Kim, S. (2023). Thrust panel. U.S. Patent **No. US11773808**, registered 2023-10-03.
3. **Lee, J.** (first inventor), Kim, S., & Lee, A. (2024). Thruster for micro-satellite and manufacturing method thereof. U.S. Patent Application No. 18/827975 (2024-09-09).

Domestic Patents (14; 6 registered + 8 pending)

i) Production method of solid projectile (KR 10-2181636, registered 2020-11-17); ii) Mass-fraction controlled solid propellant (KR 10-2199780, registered 2020-12-31); iii) Combustion gas recirculation device (KR 10-2302860, registered 2021-09-10); iv) Thrust panel (KR 10-2413082, registered 2022-06-21); v) Manufacturing method for high-heat-resistant nozzles (KR 10-2435063, registered 2022-08-17); vi) Solar panel module and spacecraft (KR 10-2496872, registered 2023-02-02); vii) Thruster panel (KR App. 10-2022-0012149, filed 2022-01-27); viii) Manufacturing method for propellant grain (KR App. 10-2022-0012721, filed 2022-01-27); ix) Dip-suction coating device and method (KR App. 10-2023-0020201, filed 2022-02-15); x) and xi) Thruster for micro-satellite and manufacturing method thereof (KR Apps. 10-2023-0123311 and 10-2023-0123312, both filed 2023-09-15); xii) Double-layer buoy with mycelium composite material (KR App. 10-2024-0003651, filed 2024-01-09); xiii) Gas generator for actuator (KR App. 10-2024-0051541, filed 2024-04-17); xiv) Gas generator for pneumatic apparatus (KR App. 10-2024-0057082, filed 2024-04-29).

RESEARCH GRANTS & FUNDING (PI / CO-PI)

1. **Sejong Science Fellowship, National Research Foundation of Korea (NRF)** Mar. 2026 – Feb. 2031
   Principal Investigator. **KRW 500,000,000**. "Multiphysics-Based Design of High Pressure Plasma-Chemical Hybrid Propulsion for High Thrust&Specific Impulse Space Exploration and Maneuvering Platforms"
2. **POSTECH Next Generation Convergence Research Pioneer Program** Mar. 2025 – Feb. 2026
   Subproject Leader. PI: Kyungtae Kim. **KRW 20,000,000**. "Development of a Hybrid Plasma Propulsion System for Small Satellites Enabled by Metal Additive Manufacturing"
3. **POSTECH Alchemist Program** Apr. 2024 – Feb. 2025
   Principal Investigator. **KRW 15,000,000**. "Verification of Key Enabling Technologies for Modular Space Architecture Using Miniature Untethered Pneumatic Pumps and Deployable Structures"
4. **POSTECH Alchemist Program** Nov. 2021 – Aug. 2022
   Principal Investigator. **KRW 20,315,000**. "Development of an Optimized Satellite Thruster for Small Satellite Constellation-Based Ultra-Low-Latency Internet Networks"
5. **SBS Foundation Research Fellowship** Sep. 2020 – Feb. 2026
   Principal Investigator. **KRW 97,000,000**. "Development of a Shell Theory-Based Integrated Structural Optimization System for Launch Vehicle Components"

RESEARCH PROJECTS (PARTICIPATION)

1. **NRF Young Researcher Program** Apr. 2024 – Feb. 2029
   Research Director: Prof. Anna Lee. KRW 1,200,000,000 (5 years). "Development of Modular Space Architecture Technology using Micro Untethered Pneumatic Pumps and Deployable Structures."
   Contribution: Overall research planning; led design and fabrication of microthruster-based untethered pneumatic pumps and integration with deployable origami structures.
2. **NRF Basic Science Research Program** Jun. 2023 – May 2024
   Research Director: Prof. Anna Lee. KRW 50,000,000. "Enhancing Small Satellite Constellation Formation with an Optimized Satellite Propulsion System for Ultra-Responsive Internet Deployment."
   Contribution: Overall research planning; developed Lab-on-PCB liquid and solid monopropellant microthrusters for the dual propulsion system and evaluated thrust performance.
3. **NRF Basic Science Research Program** Jun. 2020 – May 2021
   Research Director: Prof. Anna Lee. KRW 50,000,000. "Development of regression rate programmable rocket fuel."
   Contribution: Overall research planning; developed dip-oscillation coating processes to fabricate multi-layer elastomer-bonded solid propellants with programmable regression rates.

INVITED TALKS

1. **Department of Mechanical Engineering, Ajou University** (Host: Prof. Jonghyun Ha) 2026
   The Path to Next-Generation Space Propulsion: From Chemical to Plasma, and Lightsail.
2. **Intergravity Inc.** (Host: Myungbo Shim, COO) 2026
   Application of Rotating Gliding Arc (RGA) to In-Space Chemical Propulsion.
3. **Department of Mechanical Engineering, Kyungpook National University** (Host: Prof. Jeongpyo Lee) 2026
   Application of Rotating Gliding Arc (RGA) to In-Space Chemical Propulsion.
4. **Korean Air** (Host: Juhyun Sim, Senior Researcher) 2025
   Application of Rotating Gliding Arc (RGA) in Propulsion: From Microthrusters to OTV Propulsion Systems.
5. **Department of Aerospace Engineering, Chosun University** (Host: Prof. Taegyu Kim) 2025
   Application of RGA in Propulsion: From Microthrusters to OTV Propulsion Systems.
6. **Korea Aerospace Research Institute (KARI)** (Host: Dr. Hyunjun Kim) 2025
   Application of Rotating Gliding Arc (RGA) in Propulsion: From Microthrusters to Ignition Systems.
7. **LINC 3.0 Mechanical Engineering Seminar, POSTECH** (Host: Prof. Jintae Kim) 2024
   Introduction of Small-Scale Chemical and Plasma Thruster for In-Space Propulsion.
8. **POSTECH Tech Review Program** 2022
   A New Paradigm for Space Robot Control: From Satellite Thrusters to Soft Robots.
9. **POSTECH Tech Review Program** 2021
   Launch Vehicles, Satellites, and Solid Mechanics.

AWARDS & HONORS

Graduate

1. **Forbes Korea 30 Under 30 — Science Category Honoree** 2026
2. POSTECHIAN Fellowship — Achievement Track, POSTECH (KRW 3,000,000) 2026
3. Outstanding Presentation Award, PRESM 2025
4. Iksung Excellence Paper Award, Department of Mechanical Engineering, POSTECH (KRW 1,000,000) 2025
5. Excellence Paper Award (2022, 2024) and Encouragement Award (2023, 2025), KAI Aerospace Paper Contest, Korea Aerospace Industries 2022–2025
6. POSTECH Alchemist Entrepreneurship Fellowship (KRW 5,000,000) 2024, 2025
7. Best Paper Award, Korean Society of Mechanical Engineers (KSME) 2023
8. **Prime Minister's Award** — Grand Prize, Challenge! K-Startup Contest, Korea Institute of Startup & Entrepreneurship Development (KISED) (1/5,420 teams; KRW 150,000,000) 2022
9. Best Paper Award, Korean Society for Precision Engineering (KSPE) 2022
10. Grand Prize, SK Hynix Startup Competition (KRW 30,000,000) 2022
11. Excellence Award, Joo-Young Chung Entrepreneurship Contest, Asan Foundation (KRW 10,000,000) 2022

Undergraduate

12. **Mueunjae Award, POSTECH (highest graduation honor)** 2020
13. **Next-Generation Engineering Leader Award — Highest Honor, National Academy of Engineering of Korea** 2019
14. **Talent Award of Korea, Ministry of Education** 2019

• 12+ undergraduate awards in research, startup competitions, and external talent programs (POSTECH) 2015–2020

SERVICE ACTIVITIES

Journal Reviewer

- npj Nanophotonics, Applied Thermal Engineering, FirePhysChem, Plasma Chemistry and Plasma Processing

Memberships

- American Physical Society (APS); Korean Society for Aeronautical and Space Sciences (KSAS); Korean Society of Propulsion Engineers; Korean Society of Mechanical Engineers (KSME); Korean Society for Precision Engineering

TEACHING & SUPERVISING EXPERIENCE

Teaching Experience

- Co-Instructor, Introduction to Rocketry (Undergraduate), *POSTECH* 2024
- Teaching Assistant, Solid Mechanics (Undergraduate), *POSTECH* 2020
- Student Mentor, Fluid Mechanics (Undergraduate), *POSTECH* 2018
- R&E Mentor — Sounding Rocket Design & Propulsion Kit, *ChungNam Samsung High School* 2018
- R&E Mentor — Rocket Principles & 3D-Printed Thruster Development, *Nonsan Daegun High School* 2017
- R&E Mentor — Sounding Rocket Design Program, *Pohang Dongji High School* 2015

Teaching Interests

Introduction to Rocketry; Introduction to Aerospace Systems; Capstone Design in Aerospace Engineering; Fluid Mechanics; Heat Transfer; Solid Mechanics; MEMS and Microfabrication.

Supervision Experience

- Chemical–plasma propulsion (RGA thrusters): Seonghyeon Kim, Gyeom Lee, Hanseong Jo 2025
- Thin-film wrinkling and large-deformation mechanics: Junsik Kim 2024
- Soft actuators and space deployable structures: Suyeong Jung, Yuseong Song 2022
- Lab-on-PCB microthrusters and modular architectures: Eunji Lee 2021

MEDIA COVERAGE

1. KBS Pohang Radio "Lively Morning Citizen Square"— Artemis 2 mission commentary. 2026
2. Air-breathing electric propulsion at atmospheric pressure (Pub #10, ASR 2026) 2026
   Links: YTN Science, Hankook Ilbo, Dong-A Science, Chosun Biz, Kyongbuk Daily.
3. Highly stretchable electrodes via wrinkle structures (Pub #15, IJMS 2026) 2026
   Links: Veritas Alpha, Electronic Times, KPI News.
4. Forbes Korea 30 Under 30 — Science Category (Award #1, 2026) 2026
   Links: Forbes Korea (Feature), Forbes Korea (Full List).
5. N_2O/CH_4 green storable bipropellant (Pub #1, AST 2026) 2026
   Links: Dong-A Science, Chosun Ilbo, Herald Economy.
6. N_2O/NH_3 green storable bipropellant (Pub #2, Fuel) 2025
   Links: Dong-A Science, Chosun Biz.
7. Ultra-light thin-walled Ti-6Al-4V microthrusters (Pub #11, VPP 2025) 2025
   Links: Yonhap News, Dong-A Science.
8. Pohang MBC "Tok-Tok Donghaein" 2024
   Featured interview "Toward Space for the Expansion of Human Frontiers". Link: YouTube.
9. Mueunjae Award — POSTECH highest graduation honor (Award #12, 2020) 2020
   Links: POSTECH Times.
10. NAEK Next-Generation Engineering Leader Award — Highest Honor (Award #13, 2019) 2019
    Links: Yonhap News, Dong-A Science, Dong-A Ilbo.
11. Talent Award of Korea (Award #14, 2019) 2019
    Links: HelloDD.

REFERENCES

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